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Sustainable water utilization and disaster mitigation system

Abstract

The present invention is a water collection and storage station system. The system comprises one or more water tanks embedded in terrain such as natural mountains for water containment, storage, and directed outflow without requiring energy-intensive pumping, while also producing electricity. A high-altitude catchment area collects water, which is filtered through zeolite beds before storage. The system includes multiple exits for water distribution, overflow management, and debris removal. Each exit is integrated with hydroelectric generators for electricity production. The system features a heliport for firefighting support, equipped with rapid-fill mechanisms for flexible water buckets, thermal cameras for fire detection, and water cannons for fire suppression. The multifunctional system provides sustainable solutions for water conservation, energy generation, and disaster management.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/728,188, which was filed on Dec. 5, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION
(1) The present invention generally relates to water collection and management systems. More specifically, the invention relates to an integrated water collection and storage station designed to address global water concerns, including water scarcity, flood mitigation, and renewable energy generation. The system comprises a water tank embedded in a natural mountain, configured to store filtered precipitation and runoff water securely. The system features a high-altitude catchment area to collect precipitation, a zeolite filtration bed for removing contaminants, and multiple outlets (i.e., piped outlets) for water distribution, overflow management, and debris removal. Integrated hydroelectric generators generate renewable electricity. The invention also includes firefighting support features, such as thermal cameras, water cannons, and a heliport for rapid emergency response. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods of manufacture.
BACKGROUND

(2) By way of background, water is a fundamental and limited resource and is essential for the survival of all living organisms. Water scarcity, flooding, and contamination are global challenges faced by individuals. Rising temperatures, prolonged droughts, extreme precipitation, and frequent forest fires are also increasingly threatening the availability and quality of water resources worldwide. Most of the precipitation currently goes unused or causes damage, such as flooding, erosion, and water contamination. Precipitation runoff from highlands and mountains often devastates communities by washing away infrastructure and depositing contaminants into watersheds.

(3) Precipitation, as one of the primary components of the water cycle, offers immense potential to meet global water, however, is not currently and efficiently harnessed and managed. Highlands and mountainous regions provide opportunity for reliable water collection. Properly capturing and storing the precipitation at its source can mitigate the need for extensive chemical treatments required for groundwater or surface water. Stored precipitation can serve as a crucial supply of clean drinking water for cities and towns, reducing dependency on aging and inefficient water systems. Individuals desire a system that can collect, save, and distribute precipitation efficiently to provide a greener, sustainable solution to global water concerns.

(4) Therefore, there exists a long-felt need in the art for a sustainable system that provides a reliable solution for global water management challenges. There is a long-felt need for a system that minimizes water wastage, reduces dependency on aging water infrastructure, and prevents the damage caused by uncontrolled precipitation and runoff. Additionally, there is a long-felt need for an integrated system capable of addressing critical issues such as water scarcity, urban runoff management, flood mitigation, and forest fire control. Furthermore, there is a long-felt need for a system that is environmentally friendly, energy-efficient, and capable of generating renewable electricity during water collection and distribution processes. More specifically, there exists a long-felt need in the art for an improved water collection and storage system that can efficiently harness precipitation from high-altitude areas. Finally, there is a long-felt need for a system that is versatile, scalable, and equipped with features to support emergency services, such as firefighting operation.

(5) The subject matter disclosed and claimed herein, in one embodiment, comprises an integrated water collection, storage, and management system designed to address global water concerns. The system includes at least one water tank embedded in a natural mountain and uses gravity to facilitate efficient water flow without the need for extensive pumping systems. The system features a high-altitude catchment area for collecting precipitation, which is filtered using a zeolite bed before storage in the tank. The system is equipped with multiple outlets (i.e., piped outlets), including a main water exit piped or directed to towns and cities, overflow exit to prevent overflow in the tank, and clean-out outlet to remove debris from the tank, each exit is integrated with hydroelectric generators to produce renewable energy. Additionally, the system includes firefighting support features, such as a heliport for water bucket refills, thermal cameras for fire detection, and water cannons for rapid fire suppression. The system will be automated online to cities and towns fire departments for staff safety.

(6) In this manner, the water collection and storage system of the present invention accomplishes all of the foregoing objectives and provides a practical, sustainable solution for managing precipitation and runoff water. The system reduces energy consumption and offers a scalable solution for water conservation. The system uses zeolite filtration which enables clean water storage, suitable for drinking and other critical uses. The system includes thermal cameras, water cannons, and a heliport for disaster management in forested and mountainous areas, wherein the cameras detect fire and cannons are used for spraying water automatically. A plurality of hydroelectric generators are used in the system to generate electricity to power system operations and supply energy to nearby towns. The system addresses water scarcity, flood prevention, renewable energy generation, and emergency response.

SUMMARY OF THE INVENTION

(7) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.
(8) The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a water collection and storage station system configured to collect, store, and manage precipitation and runoff water. The system comprises at least one water tank embedded in a natural mountain, the water tank is configured to store water, a catchment area is disposed at a high altitude for collecting precipitation, an inlet is configured to transfer filtered water from the catchment area to the water tank using gravity, a main water exit (i.e., piped exit) is configured to transfer water from the water tank to external locations using gravity, and a ventilation boundary is disposed around the water tank, the ventilation boundary facilitates airflow and providing access for maintenance. The system helps in addressing water concerns by collecting, saving, and efficiently managing precipitation and runoff water.
(9) In another aspect, the water collection and storage station system is configured to support firefighting operations. For firefighting operations, the system includes at least one water tank embedded in a natural mountain, a heliport is disposed on or near the water tank for enabling helicopter landing and water bucket filling, a plurality of thermal cameras are disposed on thermal towers for detecting fire locations and intensities, water cannons are positioned around the system, and the water cannons are configured to spray water at high pressure for fire suppression.
(10) In another embodiment, the catchment area is positioned at a high altitude to collect precipitation directly from the atmosphere. A zeolite filtered bed is disposed between the catchment area and the water tank for removing impurities from the collected precipitation. A tank door is configured to open and close the water tank, enabling filtered water entry and a flush exit is configured to clean the catchment area by removing contaminants before enabling precipitation to enter the water tank.
(11) In one embodiment, a water collection and storage station system configured to manage urban precipitation runoff is disclosed. The system comprises at least one water tank, the water tank has at least a main water exit (i.e., piped exit) for towns, an overflow exit for preventing overflow in the tank, and a clean-out outlet for cleaning debris from the tank, a highway or road is connected to a plurality of inlets in the form of gutters or ditches, the inlets are configured to direct precipitation runoff into the water tank, a catchment area configured to collect additional precipitation runoff from urban areas, and a channel connects the water tank to urban distribution systems for providing stored water to cities and towns during droughts or water shortages. A plurality of generators are used in the system for generating electricity from water of the tank.
(12) In another aspect, a ventilation screen is included in the system and the screen is made from corrosion-resistant stainless-steel mesh and is configured to provide 60-80% airflow for optimal ventilation.
(13) In yet another aspect, the ventilation boundary includes adjustable openings to regulate airflow based on environmental conditions and enables automated adjustments to maintain optimal tank integrity and air quality.
(14) In still another embodiment, for generating electricity, a hydroelectric generator is disposed in the main water exit (i.e., piped exit) for generating electricity as water exits the tank, a second hydroelectric generator is disposed in the clean-out outlet for generating electricity during debris removal, and a third hydroelectric generator disposed in the overflow exit for generating electricity during periods of excess precipitation.
(15) Numerous benefits and advantages of this invention will become apparent to those skilled in the art to which it pertains upon reading and understanding of the following detailed specification.
(16) To the accomplishment of the foregoing and related ends, certain illustrative aspects of the

disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

(2) FIG. **1** illustrates a perspective view of one potential embodiment of water collection and storage station system of the present invention in accordance with the disclosed architecture;

(3) FIG. **2** illustrates a side perspective view of a water tank top door to enable water to filter through a zeolite filtered bed before entering the tank or water station in accordance with one embodiment of the present invention;

(4) FIG. **3** illustrates a perspective view showing of another embodiment of the water station system of the present invention in accordance with the disclosed structure;

(5) FIG. **4** illustrates a top view of the water station used in the system of the present invention in accordance with the disclosed structure;

(6) FIG. **5** illustrates a perspective view of the water station of the present invention with a highway or a road being used as water catchment to direct precipitation runoff into the system;

(7) FIG. **6** illustrates the water station system of the present invention being used with mountainous terrain in accordance with one embodiment of the present invention; and

(8) FIG. **7** illustrates a side perspective view of a flat mountain top with the integrated water station system in accordance with the disclosed structure.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

(9) The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

(10) As noted above, there exists a long-felt need in the art for a sustainable system that provides a reliable solution for global water management challenges. There is a long-felt need for a system that minimizes water wastage, reduces dependency on aging water infrastructure, and prevents the damage caused by uncontrolled precipitation and runoff. Additionally, there is a long-felt need for an integrated system capable of addressing critical issues such as water scarcity, urban runoff management, flood mitigation, and forest fire control. Furthermore, there is a long-felt need for a system that is environmentally friendly, energy-efficient, and capable of generating renewable electricity during water collection and distribution processes. More specifically, there exists a long-felt need in the art for an improved water collection and storage system that can efficiently harness precipitation from high-altitude areas. Finally, there is a long-felt need for a system that is versatile, scalable, and equipped with features to support emergency services, such as firefighting operation.

(11) The present invention, in one exemplary embodiment, is a water collection and storage station system configured to collect, store, and manage precipitation and runoff water. The system comprises at least one water tank embedded in a natural mountain, the water tank is configured to store water, a catchment area is disposed at a high altitude for collecting precipitation, an inlet is configured to transfer filtered water from the catchment area to the water tank using gravity, a main water exit (i.e., piped exit) is configured to transfer water from the water tank to external locations using gravity, and a ventilation boundary is disposed around the water tank, the ventilation boundary facilitates airflow and providing access for maintenance. The system helps in addressing water concerns by collecting, saving, and efficiently managing precipitation and runoff water.
(12) Reference will now be made in detail to the present preferred embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals are used in the drawings and the description to refer to the same or like parts.
(13) Referring initially to the drawings, FIG. **1** illustrates a perspective view of one potential embodiment of water collection and storage station system of the present invention in accordance with the disclosed architecture. The water collection and storage station system **100** of the present invention is designed to address global water concerns by collecting, saving, and efficiently managing precipitation and runoff water. More specifically, the system **100** is configured to collect precipitation, prevent runoff damage, generate electricity, and help in firefighting. As illustrated, the system **100** includes at least one water tank **102** which is embedded in one or more natural mountain **104**. The tank **102** is configured to store water and the mountains **104** function as natural high-altitude platforms for gravity-fed water system **100**. The mountains **104** also provide space for embedding water tanks **102** and gravity is used to facilitate water flow without the need for extensive pumping systems. The water tank **102** preferably stores filtered water securely for distribution to cities, towns, or firefighting efforts. The water tank **102** is equipped with a plurality of exits for overflow, cleaning, and main water distribution as described later in the disclosure.
(14) A catchment area **106** is also disposed in the system **100** and may be embedded in the mountains **104**. The catchment area **106** is configured to collect rainfall and precipitation directly from the atmosphere, and/or snowmelt from the surrounding area. The catchment area **106** is preferably positioned at high or elevated altitudes to avoid contamination from the ground.
(15) A ventilation boundary **108** also referred to as air and workspace is disposed around the tank **102** for providing ventilation around the water tank **102**. The ventilation boundary **108** also facilitates easy access for maintenance and repairs of the corresponding water tank. The boundary **108** can be made of any durable and corrosion-resistant material and helps in maintaining integrity and shape of the tank **102**. The water for storage in the tank **102** is received from the catchment area **106** via an inlet **110**. The inlet **110** forms the entry point for filtered water from the catchment area **106** into the tank **102**. The inlet **110** does not require any additional energy or pumping system for pumping the water from the catchment area **106** to the tank **102**.
(16) One advantage of the system **100** is that the water stored in the tank **102** can be transferred directly to cities and towns during droughts and to overcome water problems. A main water exit (i.e., main pipe exit) **112** is used for directing or piping stored water to cities, towns, and other areas requiring water supply (i.e., potable water). At least one channel and/or pipe **114** is used for transferring the water to the cities and towns and gravity is preferably used to transport water efficiently, reducing the need for pumping. In many cases, due to excessive rain or precipitation, the water tank **102** may overflow and for prevention of overflow of the tank **102**, excess water can be redirected to one or more secondary tanks (not shown) via an overflow exit **116**. The overflow exit **116** helps manage heavy precipitation in the catchment area **106**.
(17) A hydroelectric generator **118** is disposed in the main water exit (i.e., main pipe exit) **112** and can be selectively and independently activated for generating electricity for transmission to the towns and cities. The electricity can also be generated for providing power to one or more electrical components used in the system **100**.

(18) A clean-out outlet **120** is disposed in the tank **102** for removing debris from the tank **102** to maintain water quality of the tank **102**. A second hydroelectric generator **122** can be disposed in the outlet **120** for generating electricity during water discharge from the clean-out outlet **120**. For cleaning the precipitation collected in the catchment area **106**, a flush exit **124** is disposed in the catchment area **106**. The exit **124** is used to clean the initial precipitation from the catchment area **106** to remove contaminants, thereby enabling only clean water (i.e., potable water) for storage in the water tank **102**.
(19) A heliport **126** is included in the system **100** and is configured to provide a landing area for helicopters used in firefighting operations. The heliport **126** can be used for filling flexible water buckets quickly to combat forest fires. An additional hydroelectric generator **128** is also disposed in the water tank **102** for generating electricity. A hydroelectric generator **130** is also disposed in the flush exit **124** for generating electricity from the precipitation.
(20) A plurality of thermal cameras **132** are disposed on thermal towers **134** and are used to detect forest fires by identifying heat sources such as flames. The thermal cameras **132** can provide real-time data on fire location and intensity and may automate activation of water cannons **136** for targeted firefighting. The water cannons **136** used in different embodiments of the present invention suppress fires by spraying large volumes of water at high pressure and can be positioned strategically around collection and storage stations for rapid deployment.
(21) A screen **137** is disposed in the system **100** for providing proper air ventilation around the tank **102** and helps in preventing the build-up of moisture and harmful gases within the system **100**. The screen **137** is preferably in the form of a mesh and is made from a corrosion-resistant and lightweight material. The screen is configured to enable at least 60-80% airflow therethrough for tank ventilation and minimizing moisture buildup.
(22) FIG. **2** illustrates a side perspective view of a water tank top door to enable water to filter through a zeolite filtered bed before entering the tank in accordance with one embodiment of the present invention. The system **100** uses at least one Zeolite filtered bed **138** for filtering water entering the tank to remove impurities and contaminants. The bed **138** is preferably made of Zeolite **140** and is preferably positioned between the catchment area **106** and the tank entrance **110**. The tank **102** also includes at least one tank door **142** for opening and closing the tank **102**.
(23) FIG. **3** illustrates a perspective view showing of another embodiment of the water station system of the present invention in accordance with the disclosed structure. As illustrated, in the present embodiment, the Zeolite bed **138** containing a plurality of Zeolites **140** is disposed in a dam **302** which can collect water from precipitation, runoff, springs, rivers, lakes, and snowmelt from mountains **104**. A dam hydroelectric generator **304** is disposed in the dam **302** and clean out water is transmitted from the dam **302** via the channel **306**. The clean-out outlet **120** is disposed in the tank **102** for removing debris from the tank **102** to maintain water quality of the tank **102** and is coupled with hydroelectric generator **122**. It should be noted that the water station of the present embodiment is designed to work with one or more dams and all other components work in the same way as mentioned earlier in FIG. **1**. The dam **302** is used to stop precipitation runoff in specific paths to prevent flooding and redirect collected runoff water into tanks for storage and later use.
(24) FIG. **4** illustrates a top view of the water station used in the system of the present invention in accordance with the disclosed structure. In the present embodiment, a plurality of rock barricades **402** are disposed in front of the catchment area **106** to prohibit rocks and to slow down floodwaters before being collected by the catchment area **106**. The rock barricades **402** are configured to stop rocks from moving into dam **302**. The rock barricades **402** also protect the Zeolite bed **138** from physical impact.
(25) FIG. **5** illustrates a perspective view of the water station of the present invention with a highway or a road being used as water catchment to direct precipitation runoff into the system. In the present embodiment, a highway or a road **502** is connected to one or more inlets **504** which can

be in the form of gutters or ditches and the collected water is transported to the tank **102**. The tank **102**, similar to FIG. **1**, has a plurality of outlets including clean-out outlet **120**, main water exit (i.e., piped exit) **112**, and overflow exit **116**. The present embodiment is useful to provide additional sources of water collection in populated areas. Further, the system helps manage urban water runoff efficiently.

(26) FIG. **6** illustrates the water station system of the present invention being used in mountainous terrain in accordance with one embodiment of the present invention. As illustrated, in the present embodiment, two tanks **602**, **604** are used and can be identical or different based on the requirements of the users. Each tank **602**, **604** receives water from the catchment area **106** and has corresponding clean-out outlet **120**, main water exit **112**, and overflow exit **116** along with respective hydroelectric generator. The system **600** is designed to optimize water collection based on the shape of the terrain and the mountain.

(27) FIG. **7** illustrates a side perspective view of a flat mountain top with the integrated water station system in accordance with the disclosed structure. In the present embodiment, the catchment area **702** has a plurality of water cannons **704** and the catchment area **702** is associated with tanks **706**, **708** via an inlet **710**. The tanks **706**, **708** have corresponding clean-out outlet **120**, main water exit **112**, and overflow exit **116** along with respective hydroelectric generator. The screen **137** is disposed in the system **100** for providing proper air ventilation around the tank **102** and helps in preventing the build-up of moisture and harmful gases within the system **100**.

(28) Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein "sustainable precipitation utilization and disaster mitigation system", "water station system", "advanced water conservation and hydroelectric generation system", "gravity-fed water storage and firefighting system", and "system" are interchangeable and refer to the sustainable precipitation utilization and disaster mitigation system **100** of the present invention.

(29) Notwithstanding the forgoing, the sustainable precipitation utilization and disaster mitigation system **100** of the present invention can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that it accomplishes the above stated objectives. One of ordinary skill in the art will appreciate that the sustainable precipitation utilization and disaster mitigation system **100** as shown in the FIGS. are for illustrative purposes only, and that many other sizes and shapes of the sustainable precipitation utilization and disaster mitigation system **100** are well within the scope of the present disclosure. Although the dimensions of the sustainable precipitation utilization and disaster mitigation system **100** are important design parameters for user convenience, the sustainable precipitation utilization and disaster mitigation system **100** may be of any size that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

(30) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

(31) What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations,

modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

Claims

1. A water collection and storage station system comprising: a catchment area for collecting the water; a plurality of water flow outlets; an inlet; and a ventilation boundary; a water storage tank embedded in a natural mountain; wherein said catchment area is embedded in the natural mountain; wherein said ventilation boundary is disposed around said water storage tank for providing ventilation around said water storage tank; wherein said inlet receives the water from said catchment area to said water storage tank; wherein said inlet has a filtering bed for filtering water from said catchment area; wherein the water is collected from the group consisting of precipitation, runoff, springs, rivers, lakes, and snowmelt from mountains; and further wherein the water is gravity fed from said catchment area to said inlet.
2. The water collection and storage station system of claim 1 further comprising a water exit.
3. The water collection and storage station system of claim 2, wherein said water exit is an overflow water exit for directing the water from said water storage tank.
4. The water collection and storage station system of claim 3, wherein said water exit has a first hydroelectric generator disposed in said water exit for generating electricity from the water passing through said first hydroelectric generator.
5. The water collection and storage station system of claim 4, wherein said water storage tank has a clean-out outlet for removing debris from said water storage tank.
6. The water collection and storage station system of claim 5 further comprising a flush exit, wherein said flush exit is between said catchment area and said water storage tank for cleaning the water collected in said catchment area.
7. The water collection and storage station system of claim 6 further comprising a heliport, wherein said heliport is proximal to said catchment area for providing a landing area for a helicopter to fill flexible water buckets in said catchment area.
8. The water collection and storage station system of claim 6, wherein said flush exit has a second hydroelectric generator for generating electricity from the water passing through said second hydroelectric generator.
9. The water collection and storage station system of claim 6, wherein the filtering bed is a zeolite filter bed positioned between said water storage tank and said catchment area for filtering the water before entering said water storage tank.